1 Project name

FoodWise

Expiry aware recipe recommendations to support food-waste prevention

2 Overview

FoodWise will help small soup kitchens and food-service teams find recipes from available ingredients and give priority to stock with approaching user-entered use-by dates. Users enter ingredient names, stock details and dates. The tool returns ranked recipes, matched ingredients and any missing items. Its primary contribution is to **SDG 12 target 12.3**. Support for SDG 2 is indirect, through more effective use of food resources.

3 Background

The United Nations reports that 1.05 billion tonnes of food were wasted at retail, food-service and household levels in 2022 [1]. Choosing recipes without considering stock or approaching dates can leave usable ingredients unused. Our project investigates whether ingredient-based NLP retrieval and explicit date prioritisation can support this decision. Actual waste reduction will require later operational measurements; the recipe dataset contains no observed waste outcomes.

4 Key objectives and research questions

Objectives. Identify recipes that fit available ingredient types; make approaching dates visible in ranking; show missing ingredients; and evaluate the first retrieval approaches transparently.

Research questions. Which approach best recovers a source recipe from a partial ingredient query? How can expiry-aware priorities be added without hiding ingredient gaps? Which source-data limitations must be resolved before portion and mass calculations?

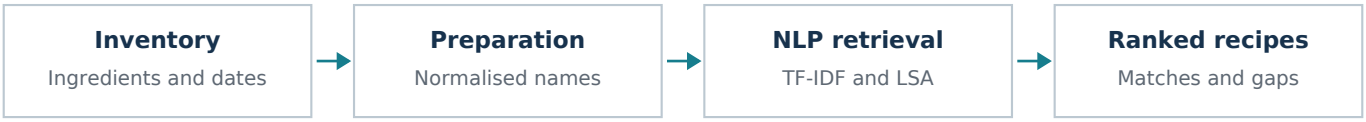
5 Methods and workflow

a Datasets

The working dataset contains exactly **8,000 real Food.com recipe records** and 11 retained fields, including ingredient names, source quantities, servings and instructions [2]. It was prepared with seed 42, and the publisher states CC0 Public Domain. All EDA and model experiments use the same 8,000 records. USDA FoodData Central is a supplementary source planned for later validated nutrient matching [5].

b Preparation and proposed tool flow

Source lists are parsed without executing their contents. Ingredient names are normalised. The sample retains named recipes with 3-20 distinct ingredients and positive known servings, with one candidate per normalised ingredient set. Stock amounts are recorded as inputs, but this stage does not infer gram-level feasibility because source quantities and units need curation.



c Exploratory data analysis

Within the 8,000-record sample, ingredient and quantity lists differ in length in **6,209 records (77.6%)**; **2,897 (36.2%)** quantity vectors contain a missing entry. The median is 8 distinct ingredients, while 470 records have source servings above 20 and need review. Positive known servings were a selection rule, so sample completeness must not be generalised to the original source.

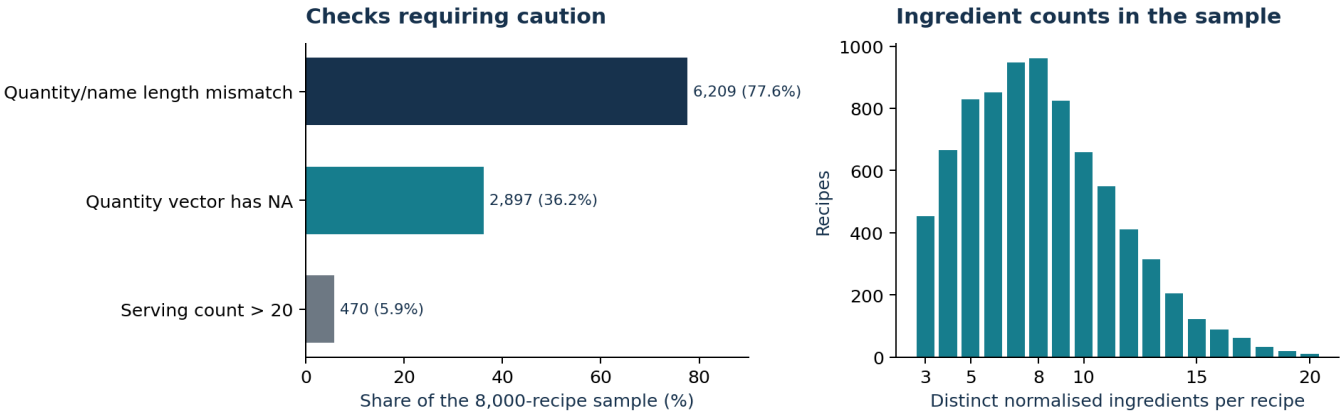


Figure 1. EDA of the 8,000-recipe working sample. Quantity flags use the source arrays; ingredient counts use distinct normalised labels.

d First model experiments and validation

The baseline uses ingredient-set overlap. TF-IDF learns ingredient weights and ranks by cosine similarity; 64-dimensional LSA tests a compact learned representation [3,4]. In 200 reproducible recipe-derived queries, 60% of each source recipe's ingredients are retained, with 3-6 query ingredients, against the 8,000-recipe index.

Approach	Recovery at 5	Recovery at 10	MRR
Ingredient overlap	100.0%	100.0%	0.974
TF-IDF cosine	99.0%	99.5%	0.954
LSA 64 dimensions	69.0%	79.0%	0.572

Simple overlap performs best on this lexical sanity check; LSA does not improve recovery. TF-IDF remains an explainable NLP candidate for further independent testing. These are **source-recipe recovery results**, not supervised test accuracy, human relevance scores or food-waste reduction. Similar alternative recipes may also be valid. The inventory demo uses explicitly simulated dates and displays ingredient gaps.

e Ethical assessment and deliverables

The tool must not infer food safety from dates alone, invent available stock or promise nutritional adequacy. Dates and usability are user inputs; entered expired or unusable stock is excluded. English ingredient vocabulary, dietary restrictions and general-recipe data limit transfer to real soup kitchens. Later validation will use independently specified pantries, user relevance judgements and curated quantity/unit mappings. Assignment 1 supplies this proposal, executed EDA notebook, initial experiments, reproducible sample and repository instructions.

Sources

[1] United Nations. SDG 12 and target 12.3. Food-waste estimate refers to 2022.

[2] irkaal. Food.com Recipes and Reviews, recipes.csv. Retrieved 17 October 2026; historical recipe publication dates 1999-2020.

[3] scikit-learn. Text feature extraction. [4] scikit-learn. TruncatedSVD. [5] USDA. FoodData Central.